

MTH203: Sequences and series

Lecture 09

1. ORDER AND LIMITS

The following theorem says that the computation of limits preserves order relations:

Theorem 1 (Order limit theorem). *Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences of real numbers. Suppose that $\lim x_n = x$ and $\lim y_n = y$. Suppose that $x_n \leq y_n$ for all n . Then $x \leq y$.*

Proof. Suppose that $y < x$. We will obtain a contradiction. Since $x - y > 0$, we choose a real number ϵ such that $0 < \epsilon < (x - y)/2$.

There exists an integer N_1 such that if $n > N_1$, we have $|x - x_n| < \epsilon$. Similarly, there exists an integer N_2 such that if $n > N_2$, then $|y - y_n| < \epsilon$. Let $N = \max\{N_1, N_2\}$.

Let $n > N$. Then $|x - x_n| < \epsilon$ and so $x - x_n < \epsilon$. Thus,

$$x_n > x - \epsilon > x - (x - y)/2 = (x + y)/2.$$

Similarly, $|y - y_n| < \epsilon$ and so $y_n - y < \epsilon$. Thus,

$$y_n < y + \epsilon < y + (x - y)/2 = (x + y)/2.$$

Thus, $y_n < (x + y)/2 < x_n$, which contradicts the assumption. This contradiction shows that we cannot have $y < x$. Thus, $x \leq y$. $\square$

Corollary 2. *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers and let c be any real number.*

(a) *If $x_n \leq c$ for all n , then $x \leq c$.*

(b) *If $x_n \geq c$ for all n , then $x \geq c$.*

Proof. Let $(y_n)_{n \in \mathbb{N}}$ be the sequence defined by $y_n = c$ for all n . Now, both statements follow if we apply the previous theorem to the sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$. $\square$

Theorem 3 (Squeeze theorem). *Let $(x_n)_{n \in \mathbb{N}}$, $(y_n)_{n \in \mathbb{N}}$ and $(z_n)_{n \in \mathbb{N}}$ be sequences of real numbers such that $x_n \leq y_n \leq z_n$ for all n . Assume that $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are convergent and $\lim x_n = \lim z_n = L$. Then, the sequence $(y_n)_{n \in \mathbb{N}}$ is also convergent and $\lim y_n = L$.*

Proof. Let $\epsilon > 0$ be given. There exists an integer N_1 such that if $n > N_1$, we have $|L - x_n| < \epsilon$. Similarly, there exists an integer N_2 such that if $n > N_2$, then $|L - z_n| < \epsilon$. Let $N = \max\{N_1, N_2\}$.

If $n > N$, we have $|L - x_n| < \epsilon$ and so $L - x_n < \epsilon$. Thus, $x_n > L - \epsilon$. Since $y_n \geq x_n$, we have $y_n > L - \epsilon$.

Similarly, we have $|L - z_n| < \epsilon$ and so $z_n - L < \epsilon$. Thus, $z_n < L + \epsilon$. Since $y_n \leq z_n$, we have $y_n < L + \epsilon$.

Thus, we see that for $n > N$, we have $y_n \in V_\epsilon(L)$. This proves that $\lim_{n \in \mathbb{N}} y_n = L$. $\square$

2. ALGEBRAIC OPERATIONS AND LIMITS

The following theorem shows that limits behave well with respect to the operation of addition:

Theorem 4. *Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences in $\mathbb{R}$. If $(z_n)_{n \in \mathbb{N}}$ is the sequence defined by $z_n := x_n + y_n$ for all n , then $(z_n)_{n \in \mathbb{N}}$ is a convergent sequence and*

$$\lim_{n \in \mathbb{N}} z_n = \lim_{n \in \mathbb{N}} x_n + \lim_{n \in \mathbb{N}} y_n.$$

Proof. Let $x = \lim_{n \in \mathbb{N}} x_n$ and let $y = \lim_{n \in \mathbb{N}} y_n$. Let $\epsilon > 0$ be any given positive real number. We want to show that there exists an integer N such that for all $n > N$, we have $|x + y - z_n| < \epsilon$.

Informal discussion: The key idea is to use the following computation:

$$\begin{aligned} |x + y - z_n| &= |x + y - (x_n + y_n)| \\ &= |(x - x_n) + (y - y_n)| \\ &\leq |x - x_n| + |y - y_n| \end{aligned}$$

We would like to show that for sufficiently large n , $|x + y - z_n|$ is smaller than ϵ . This is achieved if we can arrange for each of the two quantities $|x - x_n|$ and $|y - y_n|$ to be less than $\epsilon/2$.]

There exists an integer N_1 such that if $n > N_1$, we have $|x - x_n| < \epsilon/2$. Similarly, there exists an integer N_2 such that if $n > N_2$, then $|y - y_n| < \epsilon/2$. Let $N = \max\{N_1, N_2\}$. Then, if $n > N$, we have both $|x - x_n| < \epsilon/2$ and $|y - y_n| < \epsilon/2$. Thus, by the above computation, it follows that $|x + y - z_n| < \epsilon$. $\square$

We also have a corresponding result for multiplication:

Theorem 5. *Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences in $\mathbb{R}$. If $(z_n)_{n \in \mathbb{N}}$ is the sequence defined by $z_n := x_n y_n$ for all n , then $(z_n)_{n \in \mathbb{N}}$ is a convergent sequence and*

$$\lim_{n \in \mathbb{N}} z_n = \left(\lim_{n \in \mathbb{N}} x_n \right) \cdot \left(\lim_{n \in \mathbb{N}} y_n \right).$$

Proof. Let $x = \lim_{n \in \mathbb{N}} x_n$ and let $y = \lim_{n \in \mathbb{N}} y_n$. Let $\epsilon > 0$ be any given positive real number. We want to show that there exists an integer N such that for all $n > N$, we have $|xy - z_n| < \epsilon$.

Informal discussion: The key idea is to use the following computation:

$$\begin{aligned} |xy - z_n| &= |xy - x_n y_n| \\ &= |xy - x y_n + x y_n - x_n y_n| \\ &= |x(y - y_n) + y_n(x - x_n)| \\ &\leq |x| \cdot |y - y_n| + |y_n| \cdot |x - x_n| \end{aligned}$$

We would like each of the quantities $|x| \cdot |y - y_n|$ and $|y_n| \cdot |x - x_n|$ to be small. Let us say that we want both of them to be smaller than $\epsilon/2$.

We already know that $|y - y_n|$ is small if n is large. This small quantity is being multiplied by the fixed number $|x|$ and so the product can be made small if $|y - y_n|$ is arranged to be small enough. For instance, we can choose n to be large enough so that $|y - y_n| < (\epsilon/2) \times (1/|x|)$. However, we will run into a problem here if $|x| = 0$. So, just to be safe, we will pick n to be large enough so that $|y - y_n| < \frac{\epsilon}{2(|x|+1)}$.

A similar idea should work for the second term. However, our problem here is that $|y_n|$ may keep changing as n increases. However, we will use the fact that the sequence $(y_n)_{n \in \mathbb{N}}$ is bounded. Thus, there exists some $M > 0$ such that $|y_n| < M$ for all n . So, if we pick N to be large enough so that $|x - x_n| < \frac{\epsilon}{2M}$, it should do the trick.

We will now write the proof formally.

Since $\lim y_n = y$, there exists an integer N_1 such that for all $n > N_1$, we have

$$|y - y_n| < \frac{\epsilon}{2 \cdot (|x| + 1)}$$

Thus, for $n > N$, we have

$$|x| \cdot |y - y_n| < |x| \cdot \frac{\epsilon}{2 \cdot (|x| + 1)} < \epsilon/2.$$

The sequence $(y_n)_{n \in \mathbb{N}}$ is bounded. Thus, there exists $M > 0$ such that for $|y_n| < M$ for all n . Since $\lim_{n \in \mathbb{N}} x_n = x$, there exists an integer N_2 such that for all $n > N_2$, we have

$$|x - x_n| < \frac{\epsilon}{2M}$$

Thus, for $n > N$, we have

$$|y_n| \cdot |x - x_n| < M \cdot \frac{\epsilon}{2M} = \epsilon/2.$$

Adding the two inequalities we have obtained, we get

$$|x| \cdot |y - y_n| + |y_n| \cdot |x - x_n| < \epsilon.$$

Combining this with the inequality

$$|xy - z_n| = |xy - x_n y_n| \leq |x| \cdot |y - y_n| + |y_n| \cdot |x - x_n|$$

(proved in the informal discussion above), we get the required result. $\square$

Finally, we deal with multiplicative inverses:

Theorem 6. Let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence in $\mathbb{R}$ such that $x_n \neq 0$ for all n and $\lim_{n \in \mathbb{N}} x_n \neq 0$. If $(y_n)_{n \in \mathbb{N}}$ is the sequence defined by $y_n := 1/x_n$ for all n , then $(y_n)_{n \in \mathbb{N}}$ is a convergent sequence and

$$\lim_{n \in \mathbb{N}} y_n = \frac{1}{\lim_{n \in \mathbb{N}} x_n}.$$

Proof. Let $x = \lim_{n \in \mathbb{N}} x_n$ and let $\epsilon > 0$ be any given positive real number. We want to show that there exists an integer N such that for all $n > N$, we have $|(1/x) - (1/x_n)| < \epsilon$.

Informal discussion: The key idea is to use the following computation:

$$\begin{aligned} |(1/x) - (1/x_n)| &= \left| \frac{x_n - x}{xx_n} \right| \\ &= \left(\frac{1}{|x|} \right) \cdot \left(\frac{|x_n - x|}{|x_n|} \right) \end{aligned}$$

We would like to show that when n is large, the quantity $\left(\frac{1}{|x|} \right) \cdot \left(\frac{|x_n - x|}{|x_n|} \right)$ is small. The number $1/|x|$ is a fixed positive number, that does not depend on x . Thus, we essentially want to ensure that when n becomes very large, the quantity $\left(\frac{|x_n - x|}{|x_n|} \right)$ becomes small. We already know that the quantity $|x_n - x|$ becomes small as n becomes large. The situation is complicated by the fact that it is divided by $|x_n|$.

What if $|x_n|$ also becomes very small as n becomes large? That would perhaps prevent the number $\left(\frac{|x_n - x|}{|x_n|} \right)$ from becoming small. Now we have to recognize that $|x_n|$ cannot become arbitrarily small as n increases. Indeed, $\lim_{n \in \mathbb{N}} x_n = x$ is non-zero. So, the sequence $\{x_n\}_{n \in \mathbb{N}}$ eventually goes arbitrarily close to a number that is away from 0. Thus, after n is very large, we can ensure that $|x_n|$ is larger than some positive number, and so $1/|x_n|$ is less than some positive number. Thus, will allow us to conclude that $|x_n - x| \cdot (1/|x_n|)$ becomes small as n becomes large.

So, the key point is to use the fact that $\lim_{n \in \mathbb{N}} x_n$ is non-zero and use this to get some kind of bound on the quantity $1/|x_n|$ for all n . More precisely, we

want to find some positive number M such that $1/|x_n| < M$ for all n . Then, if $|x_n - x| \cdot (1/|x_n|) < M \cdot |x_n - x|$. So, if we want to ensure that

$$\left(\frac{1}{|x|}\right) \cdot \left(\frac{|x_n - x|}{|x_n|}\right) < \epsilon,$$

it will suffice to ensure that

$$\left(\frac{1}{|x|}\right) \cdot M \cdot |x_n - x| < \epsilon,$$

which is equivalent to

$$|x_n - x| < \frac{|x| \cdot \epsilon}{M}.$$

This can always be arranged by making n sufficiently large.

We will now write this out formally.

Since $x \neq 0$, we have $|x| > 0$ and so $|x|/2 > 0$. Since $\lim_{n \rightarrow \infty} x_n = x$, there exists N_1 such that if $n > N_1$, we have $|x - x_n| < |x|/2$. (Thus, when n is large enough, x_n stays “sufficiently close” to x and hence “sufficiently away” from 0.) Since $|x_n| + |x - x_n| \geq |x|$, we see that $|x_n| \geq |x| - |x - x_n|$. Thus, when $n > N_1$, we have $|x_n| \geq |x| - |x|/2 = |x|/2$. Thus, when $n > N_1$, we have $1/|x_n| < 2/|x|$.

Let $M = 1 + \max\left(\{1/|x_1|, 1/|x_2|, \dots, 1/|x_{N_1}| \} \cup \{2/|x|\}\right)$. Then, we can see that $1/|x_n| < M$ for all $n \in \mathbb{N}$. This is because, if $n < N_1$, we have

$$1/|x_n| \leq \max\{1/|x_1|, 1/|x_2|, \dots, 1/|x_{N_1}| \} < M,$$

and if $n > N_1$, then we have

$$1/|x_n| < 2/|x| < M.$$

Thus, we have $1/|x_n| < M$ in either case.

Now, since $\lim_{n \rightarrow \infty} x_n = x$, there exists an integer N such that if $n > N$, we have $|x_n - x| < \frac{|x| \cdot \epsilon}{M}$. Thus, if $n > N$, we have

$$M \cdot \left(\frac{|x_n - x|}{|x_n|}\right) < \epsilon.$$

Combining this with the inequality

$$\frac{|x_n - x|}{|x_n x|} = \left(\frac{1}{|x_n|}\right) \cdot \frac{|x_n - x|}{|x|} < M \cdot \left(\frac{|x_n - x|}{|x|}\right),$$

we obtain the inequality

$$\frac{|x_n - x|}{|x_n x|} < \epsilon$$

for $n > N$. This proves the result. $\square$

This gives us the following result:

Corollary 7. *Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences in $\mathbb{R}$. Assume that $y_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim_{n \in \mathbb{N}} y_n \neq 0$.*

If $(z_n)_{n \in \mathbb{N}}$ is the sequence defined by $z_n := x_n/y_n$ for all n , then $(z_n)_{n \in \mathbb{N}}$ is a convergent sequence and

$$\lim_{n \in \mathbb{N}} z_n = \frac{\lim_{n \in \mathbb{N}} x_n}{\lim_{n \in \mathbb{N}} y_n}.$$

Proof. This corollary is an immediate consequence of Theorems 5 and 6. $\square$

3. READING SUGGESTIONS

This lecture corresponds to part of Section 2.3 of the textbook ([1]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [5](#)